Physics LLM Presentation Pack

Assembled Agent Prompts

12 tasks • 15 edges • Exported March 11, 2026

> Each card below shows the **full assembled prompt** that Astro's dispatch engine sends to the agent for each task. This mirrors `ContextAssembler.buildExecutionPrompt()` — the sections are assembled at dispatch time from the project vision document, dependency graph, completed task outputs, and the task description. Sections not available at export time (skill instructions, runtime knowledge) are omitted.

Vision | Context | Dependencies | Task (prompt) | Instructions

Tasks

- #1 Bootstrap ArXiv LaTeX Workspace
- #2 Paper Structure and Contribution Extraction
- #3 Figure Inventory and Visual Asset Plan
- #4 Narrative Blueprint and 40-Slide Storyboard
- #5 Generate Slides 01-12 (Hook + Background)
- #6 Generate Slides 13-28 (Method + Core Evidence)
- #7 Generate Slides 29-40 (Extensions + Discussion)
- #8 Assemble and Render Dual-Theme Slide PDFs
- #9 Generate Detailed Blog Post
- #10 Generate Portrait Poster and A0 PDF
- #11 Cross-Deliverable QA and Consistency Gate
- ◆ #12 Presentation Pack Complete

#1 Bootstrap ArXiv LaTeX Workspace

PROJECT VISION

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TASK CONTEXT

```
15 tasks depend on this one.
```

CURRENT TASK

```
**Bootstrap ArXiv LaTeX Workspace**

Initialize the project from arXiv first: download and extract https://arxiv.org/e-
print/2305.13673 into `source/arxiv-2305.13673/`, then create working folders `outline/`,
`assets/figures/`, `assets/svg/`, `output/slides/`, `output/blog/`, `output/poster/`. Do
not start content generation yet. Done means the LaTeX source (including `cfg-ob4.tex` and
figure assets) is present and directory scaffolding is ready.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#2 Paper Structure and Contribution Extraction

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 1 in the plan graph. It has 1 upstream dependencies. 14 tasks depend on
this one.
```

CURRENT TASK

```
**Paper Structure and Contribution Extraction**

Parse `source/arxiv-2305.13673/cfg-ob4.tex` and extract: section hierarchy, key equations,
bibliography signals, and all `\includegraphics` references. Write `outline/paper-
structure.md` with 3-5 key contributions and evidence anchors (section/equation/figure
refs). If a real GitHub URL is provided (not `{{githubUrl}}`), clone into `external/repo/`
and add `outline/repo-notes.md` identifying key code files; otherwise record repo as
unavailable in `outline/repo-notes.md`.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#3 Figure Inventory and Visual Asset Plan

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 2 in the plan graph. It has 2 upstream dependencies. 13 tasks depend on
this one.
```

CURRENT TASK

```
**Figure Inventory and Visual Asset Plan**

Create `outline/figure-inventory.md` by mapping paper figures to presentation needs using
the rubric (original vs original+annotated vs new SVG). Require a 40-slide-ready plan with
at least 22 visuals and >60% content slides visualized. Include per-slide visual tags like
`[original: path]`, `[SVG: description]`, or `[none]`. Prioritize original experimental
figures and define beginner-focused new SVGs for Transformer basics and CFG parsing
examples.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#4 Narrative Blueprint and 40-Slide Storyboard

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 3 in the plan graph. It has 3 upstream dependencies. 12 tasks depend on
this one.
```

CURRENT TASK

```
**Narrative Blueprint and 40-Slide Storyboard**

Write `outline/narrative-outline.md` with a 500+ word comprehension summary and full 40-
slide storyboard following Motivation → Methodology → Results → Discussion. For each slide
specify title, layout type (1-5), 3-5 key points, equation plan, figure plan, and
transition line. Must include beginner-friendly but concise background section on
Transformer setup and CFG concepts with concrete examples/visualization specs.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#5 Generate Slides 01-12 (Hook + Background)

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 4 in the plan graph. It has 4 upstream dependencies. 3 tasks depend on
this one. This task is on an exploration branch. 4 other tasks are running in parallel.
```

CURRENT TASK

```
**Generate Slides 01-12 (Hook + Background)**

Create `output/slides/slide_01.html` through `slide_12.html` using the 5 required layouts
and palette/typography rules. Focus on title, motivation, prior gap, and beginner-friendly
foundations (Transformer architecture, context-free grammar intuition, concrete parsing
examples). Add high-quality custom SVGs for prerequisite concepts and early pipeline
visuals.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#6 Generate Slides 13-28 (Method + Core Evidence)

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 4 in the plan graph. It has 4 upstream dependencies. 3 tasks depend on
this one. This task is on an exploration branch. 4 other tasks are running in parallel.
```

CURRENT TASK

```
**Generate Slides 13-28 (Method + Core Evidence)**

Create `output/slides/slide_13.html` through `slide_28.html` covering CFG family design,
model behavior interpretation, attention pattern findings, and dynamic-programming
connection. Reuse many original paper figures (authoritative results/plots) and add SVG
simplifications where paper visuals are too dense. Ensure equations are annotated in plain
language.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#7 Generate Slides 29-40 (Extensions + Discussion)

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 4 in the plan graph. It has 4 upstream dependencies. 3 tasks depend on
this one. This task is on an exploration branch. 4 other tasks are running in parallel.
```

CURRENT TASK

```
**Generate Slides 29-40 (Extensions + Discussion)**

Create `output/slides/slide_29.html` through `slide_40.html` for robustness, implicit CFGs,
PTB/other CFG extensions, uniform-attention findings, limitations, implications, and future
directions. Keep researcher-level detail while preserving beginner clarity through recap
visuals and concise takeaways.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#8 Assemble and Render Dual-Theme Slide PDFs

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 5 in the plan graph. It has 9 upstream dependencies. 2 tasks depend on
this one. This task is on an exploration branch.
```

CURRENT TASK

```
**Assemble and Render Dual-Theme Slide PDFs**

Combine all 40 slide HTML files into print-ready combined HTML and render PDFs for both
themes: `output/slides/slides-light.pdf` and `output/slides/slides-dark.pdf` (1280x720
pages). Ensure no clipping/overflow and preserve figure legibility. Keep source HTML self-
contained and consistent with the skill styling spec.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#9 Generate Detailed Blog Post

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 4 in the plan graph. It has 4 upstream dependencies. 2 tasks depend on
this one. This task is on an exploration branch. 4 other tasks are running in parallel.
```

CURRENT TASK

```
**Generate Detailed Blog Post**

Write `output/blog/post.md` (2000-4000 words) with TL;DR, Motivation, Approach, Key
Results, Discussion, References. Include equation blocks with plain-English explanation,
quantitative tables, and all figures used in slides (original + new SVGs) with standalone
captions. Keep terminology aligned with slide deck.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#10 Generate Portrait Poster and A0 PDF

PROJECT VISION

▶ 47,429 chars — click to expand

TASK CONTEXT

```
This task is at depth 4 in the plan graph. It has 4 upstream dependencies. 2 tasks depend on
this one. This task is on an exploration branch. 4 other tasks are running in parallel.
```

CURRENT TASK

```
**Generate Portrait Poster and A0 PDF**

Create `output/poster/poster.html` using portrait A0 3-column CSS grid and render
`output/poster/poster.pdf`. Include header, motivation/background, method, key insight,
results, discussion, limitations, and QR/citation block. Enforce column balancing (>=95%
fill per column) with real content (figures/callouts), not stretched whitespace.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

#11 Cross-Deliverable QA and Consistency Gate

PROJECT VISION

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TASK CONTEXT

```
This task is at depth 6 in the plan graph. It has 14 upstream dependencies. 1 tasks depend on
this one. This task is on an exploration branch.
```

CURRENT TASK

```
**Cross-Deliverable QA and Consistency Gate**

Run final quality checks across slides, blog, and poster: narrative consistency, claim-
number consistency, equation explanation coverage, figure coverage targets, citation
completeness, and rendering sanity checks. Produce `output/qa/report.md` with pass/fail
checklist and required fixes if any.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```

◆ #12 Presentation Pack Complete

PROJECT VISION

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TASK CONTEXT

```
This task is at depth 7 in the plan graph. It has 15 upstream dependencies. This task is on an
exploration branch.
```

CURRENT TASK

```
**Presentation Pack Complete**

Slides (light+dark PDFs), detailed blog post, and portrait poster are complete and quality-
reviewed.
```

INSTRUCTIONS

```
Execute the task described above. Focus on completing the task thoroughly and correctly.
- Use appropriate tools to accomplish the task
- Report progress and any issues encountered
- Verify your work when possible
- Be thorough but efficient
- When you make code changes, commit them with a descriptive commit message before finishing.
Do not leave uncommitted changes.
```